



DDX20 positively regulates the interferon pathway to inhibit viral infection

Zhiqiang Chen^{a,b,1}, Jinyu Zhang^{b,1}, Tingting Feng^{b,c,1}, Xiujuan Wang^b, Shimeng Zhou^b, Wen Pan^b, Zhengrong Chen^{d,**}, Yongdong Yan^{d,***}, Jianfeng Dai^{b,*}

^a Department of Nuclear Medicine, The First Affiliated Hospital of Soochow University, Suzhou, China

^b Institute of Biology and Medical Sciences, MOE Key Laboratory of Geriatric Diseases and Immunology, Jiangsu Key Laboratory of Infection and Immunity, Soochow University, Suzhou, China

^c Central Laboratory, The Fourth Affiliated Hospital of Nanjing Medical University, Nanjing, China

^d Department of Respiratory Medicine, Children's Hospital of Soochow University, Suzhou, China

ARTICLE INFO

Keywords:

DDX20
IFN- β
Innate immunity
RNA helicase
Antiviral function

ABSTRACT

The DEAD-box (DDX) family comprises RNA helicases characterized by the conserved sequence D(Asp)-E(Glu)-A(Ala)-D(Asp), participating in various RNA metabolism processes. Some DDX family members have been identified for their crucial roles in viral infections. In this study, RNAi library screening of the DDX family unveiled the antiviral activity of DDX20. Knockdown of DDX20 enhanced the replication of viruses such as vesicular stomatitis virus (VSV) and herpes simplex virus type I (HSV-1), while overexpression of DDX20 significantly diminished the replication level of these viruses. Mechanistically, DDX20 elevated the phosphorylation level of IRF3 induced by external stimuli by facilitating the interaction between TBK1 and IRF3, thereby promoting the expression of IFN- β . The increased IFN- β production, in turn, upregulated the expression of interferon-stimulated genes (ISGs), including Cig5 and IFIT1, thereby exerting the antiviral effect. Finally, in an *in vivo* infection study, *Ddx20* gene-deficient mice exhibited increased susceptibility to viral infection. This study provides new evidence that DDX20 positively modulates the interferon pathway and restricts viral infection.

1. Introduction

Innate immunity constitutes the organism's primary defense mechanism against microbial pathogens, wherein pattern recognition receptors (PRRs) act as the sensors of the viral genome or its replication intermediates within host cells. For example, toll-like receptors (TLRs) such as TLR3 and TLR7/8, predominantly located on endosomal membranes, detect double-stranded (ds) and single-stranded (ss) RNA from invading viruses (Alexopoulou et al., 2001; Diebold et al., 2004; Heil et al., 2004), respectively. Retinoic acid-inducible gene-I (RIG-I)-like receptors (RLRs) RIG-I and MDA5 sense cytoplasmic viral RNA and engage with MAVS (Kato et al., 2006), while cyclic GMP-AMP synthase (cGAS) recognizes nonself DNA and subsequently activates the stimulator of interferon genes (STING) (Zhong et al., 2008). These events prompt the activation of TANK binding kinase-1 (TBK1) and interferon

regulatory factor 3 (IRF3), leading to the induction of Type I interferon (IFN-I)-mediated host innate immunity against viral infections (Anwar et al., 2021).

DEAD-box (DDX) proteins, a prominent subfamily of RNA helicases, are distinguished by the conserved motif Asp-Glu-Ala-Asp (DEAD) (Ali, 2021b). These motifs facilitate multifaceted ATP-dependent conformational changes, including RNA processing, RNA unwinding, and RNPase activity. Given their distribution patterns, members of the DDX protein family are implicated in various cellular processes such as embryogenesis, spermatogenesis, and cellular growth and division (Ali, 2021a). Over recent decades, DDX RNA helicases have emerged as key participants in viral replication, exerting diverse roles in modulating the host innate immune response. For instance, DDX58 (also known as RIG-I) recruits MAVS, thereby initiating the phosphorylation of IRF-3 and IRF-7 (Kato et al., 2006), while DDX60 activates RIG-I- and

* Corresponding author. Institutes of Biology and Medical Sciences, Soochow University, Building 703, 199 Ren-ai Road, Suzhou, 215123, China.

** Corresponding author. Department of Respiratory Medicine, Children's Hospital of Soochow University, No. 303 Jing De Road, Suzhou, 215003, China.

*** Corresponding author. Department of Respiratory Medicine, Children's Hospital of Soochow University, No. 303 Jing De Road, Suzhou, 215003, China.

E-mail addresses: jackiang@163.com (Z. Chen), jinyumiss13@163.com (J. Zhang), ttfeng@suda.edu.cn (T. Feng), xiujuanwang93@163.com (X. Wang), shimengzhou98@163.com (S. Zhou), panwen88@suda.edu.cn (W. Pan), chen_zheng_rong@163.com (Z. Chen), yyd3060@126.com (Y. Yan), daijianfeng@suda.edu.cn (J. Dai).

¹ These three authors contribute equally.

MDA5-dependent IFN pathways, leading to IFN gene expression (Miyashita et al., 2011). DDX1 acts as a potential coactivator in transmissible gastroenteritis virus (TGEV)-induced IFN- β response (Zhou et al., 2017). DDX3 interacts with RLRs and RIG-I adaptor MAVS, participating in TBK1/IKK ϵ -mediated IRF activation and *Inf*b promoter induction (Oshiumi et al., 2010; Schroder et al., 2008). Furthermore, DDX4 positively regulates IFN-I-mediated antiviral activity through SOCS1 degradation (Miao et al., 2024). In contrast, DDX5 and DDX19 facilitate TBK1 degradation, thereby impeding IRF3 activation and subsequent IFN responses (Zhang et al., 2019, 2023). Similarly, DDX25 interrupts IRF3 and NF- κ B activation and blocks IFN- β production, thereby promoting RNA virus infections such as dengue virus (DENV) and vesicular stomatitis virus (VSV) (Feng et al., 2017). Better understanding the interaction of PRR pathways and DDX members is crucial for deciphering antiviral immune mechanisms and developing new strategies for combating viral infections.

Although traditionally associated with specific processes, emerging evidence suggests that several DEAD-box proteins participate in multiple cellular functions, occasionally extending beyond RNA-related activities (Fuller-Pace, 2006). DDX20, also known as DP103 or Gemin3, was initially identified as a host factor interacting with the Epstein-Barr virus (EBV) nuclear proteins EBNA2 and EBNA3C (Grundhoff et al., 1999). DDX20 governs various biological processes, including transcription, mRNA splicing, and translation. Moreover, its interaction with EBV nuclear proteins contributes to the stabilization of DDX20 protein, thereby impeding p53-mediated apoptosis (Cai et al., 2011). However, the precise role of DDX20 during virus infection remains largely unexplored.

In the current investigation, we conducted a comprehensive screening of 37 genes within the DEAD-box family using the RNAi approach to discern host factors influencing VSV infection. Notably, DDX20 emerged as a pivotal regulator in inhibiting VSV infection. *Ddx20*-deficient mice exhibited increased weight loss and decreased survival upon VSV infection. Our investigation provides significant insights into the role of DDX20 in viral infection, highlighting its capacity to hinder viral replication through modulation of the IFN pathway.

2. Materials and methods

2.1. Cellular culture and transfections

The HEK293T, HeLa, and Vero E6 cells were cultured in Dulbecco's Modified Eagle Medium supplemented with 10% fetal bovine serum and 1% penicillin/streptomycin. Cell cultures were maintained at 37 °C in a 5% CO₂ laboratory incubator, which was routinely cleaned and decontaminated.

Transfections of cells with plasmid DNA and siRNA were carried out using jetPEI (Polyplus, NY, USA) according to the manufacturer's manual. The siRNA library (RiboBio Co., China) targeting 37 genes of the DDX family was described in our previous study (Feng et al., 2017; Li et al., 2015).

2.2. Mice

Six to eight-week-old adult wild-type C57BL/6 mice were purchased from the experimental animal center of the Chinese Academy of Science (Shanghai, China).

DDX20-deficient embryos with a pure heterozygous genotype are not viable, resulting in lethality (Mouillet et al., 2008). On this basis, *Ddx20* Gene Systemic Knockout Heterozygous Mice *Ddx20*-KO First was constructed from the Cambridge-SU Genome Resource Center (CAM-SU GRC) at Soochow University.

The mice were housed in a specific pathogen-free barrier environment. All infection experiments were conducted in a BSL-2 animal facility in accordance with the guidelines of the National Animal Care and Use Committee. The Institutional Laboratory Animal Care and Use

Committee at Soochow University approved the animal studies.

2.3. Virus and infection

GFP-engineered vesicular stomatitis virus (VSV-GFP), VSV and HSV-1 were propagated and titrated on Vero E6 cells. Sendai virus (SeV) was propagated in 10-day-old embryonated chicken eggs (Beijing Laboratory Animal Research Center, Beijing, China) and titrated using a hemagglutination assay with chicken red blood cells (BeNa Culture Collection, Beijing, China). HEK293T cells were infected with VSV, HSV-1, or SeV at a multiplicity of infection (MOI) of 1 unless otherwise specified.

2.4. Real-time PCR

The mRNA expression levels of VSV-G, human genes *DDX20*, *IFN β* , *IFIT1*, *Cig5*, and mouse genes *Irfn β* , were quantified through SYBR Green-based quantitative PCR (qRT-PCR) using gene-specific primers and then normalized to a beta-actin gene. The qRT-PCR primer sequences for the genes are provided in Table S1.

2.5. Plasmid

Expression plasmids for the active caspase recruitment domain (CARD) containing form of RIG-I (RIG-I-N), MAVS, TBK1, IKK ϵ , the constitutively active IRF3 (IRF3-5D), and the luciferase reporter plasmid pIFN β -Luc (p125-Luc) were generously provided by Dr. Rongtuan Lin from McGill University, Canada (Zhao et al., 2007). The plasmids pNF- κ B-Luc, pISRE-Luc, and pRL-TK were purchased from Clontech (Mountain View, CA). Recombinant plasmids of DDX20 and shRNA-DDX20 were obtained from GENECHM Co., China. The shRNA sequences targeting the human DDX20 gene were 5'-CCGCTTCTCATTCATATTATT-3' and 5'-CGCCTGTGGATGATCG-TATTT-3'. The RNAi efficiency of specific genes was confirmed using qRT-PCR.

2.6. Luciferase reporter assays

HEK293T cells at 70%–80% confluency were transfected with 100 ng of NF- κ B, ISRE, or IFN- β luciferase reporter plasmids (firefly luciferase; experimental reporter), 10 ng of pRL-TK reporter plasmid (Renilla luciferase; internal control), and 100 ng of recombinant-expressing plasmids (vector or DDX20-FLAG/shDDX20). The cells were infected with VSV at 24 h post-transfection. Luciferase activity was determined after another 16 h using a Promega Dual-Glo kit following the manufacturer's instructions.

2.7. TCID₅₀ assay and viral growth kinetics

The cell-free supernatants were collected, and the titers of VSV were determined using a median tissue culture infective dose (TCID₅₀) assay following standard protocols on Vero cells. In brief, samples were serially diluted and inoculated into Vero cells in 96-well plates. After a 5-day incubation, cells were examined for cytopathic effects (CPE) under a light microscope. The virus titer (TCID₅₀/ml) was calculated using the Reed-Muench method.

2.8. Immunoprecipitation (IP) assays and western blot

5 × 10⁶ HEK293T cells were transfected with plasmids over-expressing or silencing DDX20 and then infected with VSV. After 16 h post-infection, cells were lysed with RIPA buffer. The lysates were incubated with Flag/HA-Ab-conjugated agarose beads (Sigma-Aldrich) at 4 °C overnight. After immunoprecipitation, proteins were subjected to SDS-PAGE and transferred onto a PVDF membrane for western blotting. The following antibodies were used for western blotting: VSV-G tag

rabbit polyclonal antibody (ABGENT, USA), DDX20 rabbit polyclonal antibody (Proteintech, USA), anti-human GAPDH polyclonal antibody (Proteintech, USA), HA-probe rabbit monoclonal antibody (CST, USA), anti-DYKDDDDK-tag mouse antibody (Cell Signaling Technology, USA), phospho-IRF-3 (Ser396) rabbit monoclonal antibody (Cell Signaling Technology, USA), IRF3 polyclonal antibody (Biolegend, USA), HRP-conjugated rabbit anti-mouse IgG mAb (Biolegend, USA). The signals were detected using an Enhanced Chemiluminescent (ECL) kit according to the manufacturer's instructions (Merck Millipore Ltd., USA).

2.9. Immunofluorescence microscopy

Plasmids encoding IRF3-HA or TBK1-FLAG were transfected into HeLa cells, with or without DDX20-FLAG/HA. At 24 h post-transfection, cells were infected with SeV at an MOI of 1. Cells were fixed in 1% paraformaldehyde and permeabilized with 1% Triton X100 at 16 h post-infection, followed by blocking with 5% FBS and incubation with primary antibodies (HA-probe rabbit monoclonal antibody and anti-

DYKDDDDK-tag mouse antibody) overnight at 4 °C. FITC-labeled Goat anti-mouse IgG (Jackson ImmunoResearch, USA) and TRITC-labeled Goat anti-rabbit IgG (H + L) (Jackson ImmunoResearch, USA) were used for incubation for 1 h. Cell nuclei were stained with DAPI. Cells were then examined using a fluorescence microscope.

2.10. Histology

Biological tissues embedded in paraffin were sectioned into 5 µm slices using a microtome. Subsequently, staining was performed with hematoxylin and eosin (H&E), and histological changes were examined under an optical microscope. Pathological alterations in pulmonary tissues were assessed based on the severity of peribronchitis, perivascular inflammation, bronchitis, alveolitis, and interstitial pneumonia, employing an adapted standard. Pathological changes in brain tissue were evaluated based on the severity of encephalitis, encompassing features such as neuronal necrosis, glial cell proliferation, neuronal saturation, mononuclear cell infiltration, and neurocyte

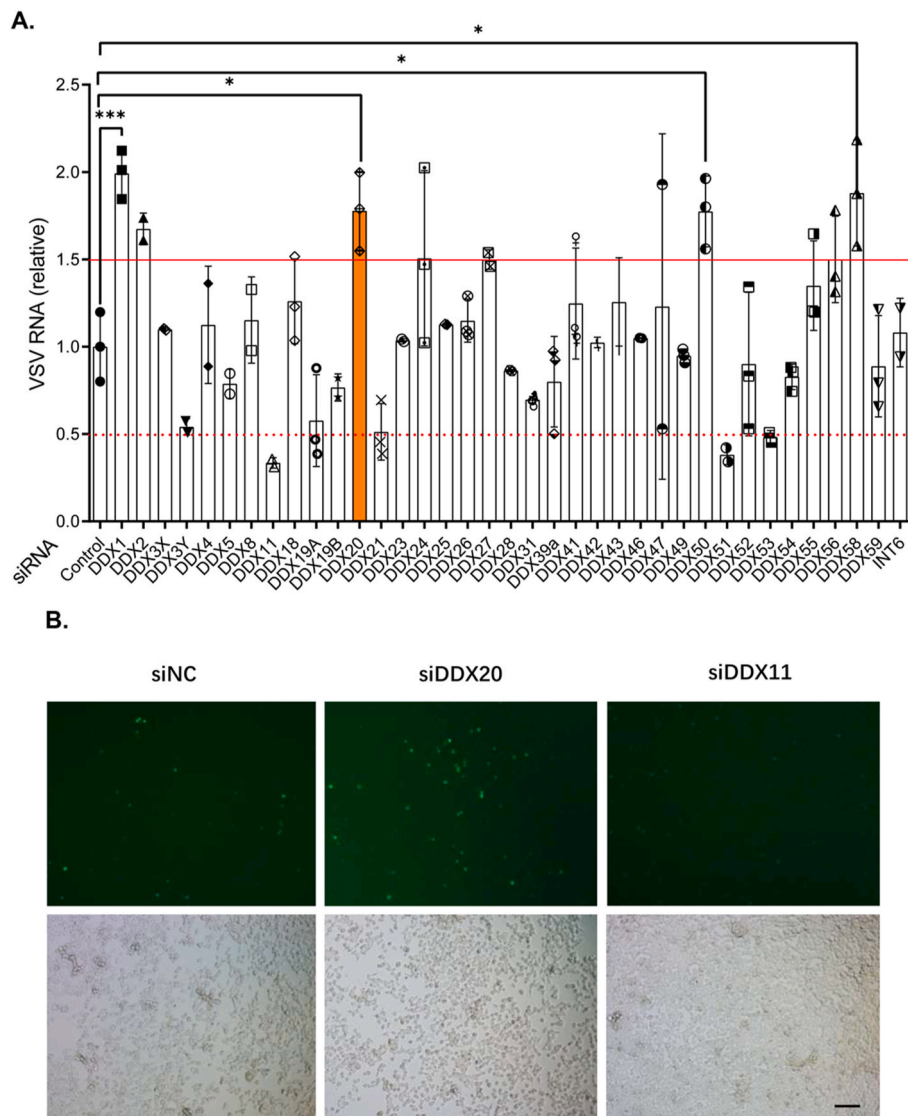


Fig. 1. Role of DEAD-box family RNA helicases during VSV infection.

(A) HEK293T cells were transfected with siRNAs targeting specific genes from the DDX RNA helicase family and infected with GFP-VSV at an MOI of 1.0. Viral loads were analyzed by measuring the GFP signaling quantify with ImageJ software. The relative replication level of VSV in nonsense siRNA (N.C.) transfected cells was set to 1.

(B) Intracellular viral loads of VSV-GFP were assessed by fluorescence microscopy. Scale bars, 100 µm.

Results are expressed as mean ± SEM. * $p < 0.05$, and *** $p < 0.001$ (t-test). Representative results are from at least three independent experiments.

phagocytosis, accompanied by lymphocytic aggregation around blood vessels.

2.11. Statistical analysis

Statistical differences in mean values were analyzed using Student's t-test and analysis of variance with GraphPad Software (PRISM 9). A p -value < 0.05 was considered statistically significant.

3. Results

3.1. Role of DEAD-box family RNA helicases during VSV infection

The DDX RNA helicases play a critical role in the antiviral innate immune response. Previous studies have established several DDX family members as sensors of viral nucleic acids or regulators of downstream type I interferon signaling. For instance, DDX58 (RIG-I) serves as a sensor for viral RNA (Loo and Gale, 2011), while cytosolic DDX3 interacts with the kinases TBK1 and IKK ϵ , thereby promoting interferon production (Schroder et al., 2008). To investigate the potential

involvement of other DDX family members in host antiviral immune responses, we conducted a screen to assess the effects of 37 DDX helicases on viral infectivity. DDX58 was included as a positive control. We achieved knockdown of their expressions using small interfering RNA (siRNA) in HEK293T cells. The siRNA screening revealed that DDX1, DDX20, DDX50, and DDX58 knockdown significantly increased the expression of VSV RNA following viral infection (Fig. 1A). Given the fact that DDX1 and DDX50 have been reported to be involved in viral infection (Han et al., 2017; Zhou et al., 2017), we hypothesized that DDX20 serves as a novel host antiviral factor during viral infection. Indeed, VSV-GFP infection experiments confirmed the heightened VSV burden in DDX20-silenced cells compared to both control and DDX11-silenced cells (Fig. 1B), suggesting a critical role for DDX20 in virus restriction.

3.2. DDX20 inhibits VSV replication

To assess the biological relevance of DDX20 in viral infection, we examined the impact of its overexpression on VSV replication. qPCR and Western blot analysis verified a significant increase in DDX20 mRNA

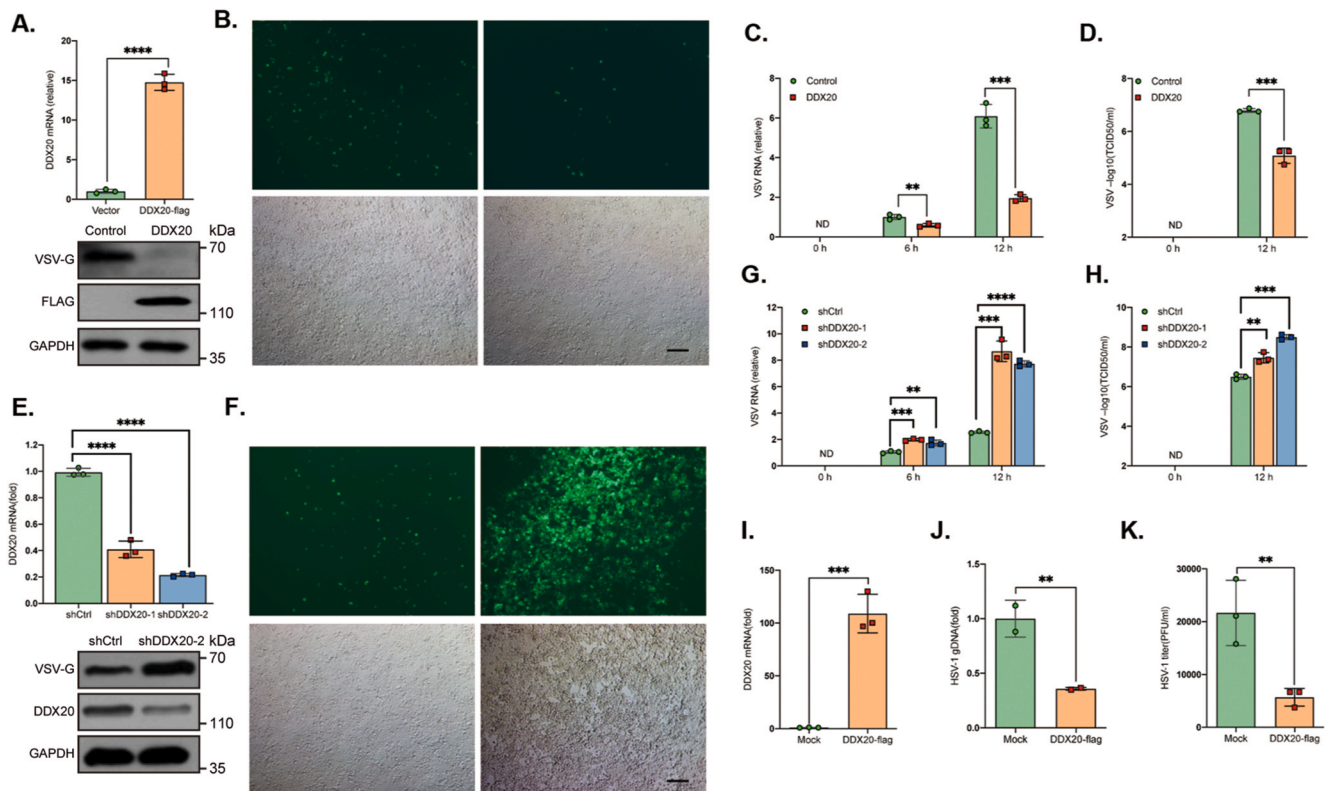


Fig. 2. DDX20 inhibits VSV replication *in vitro*.

(A) DDX20 mRNA levels were increased in human HEK293T cells upon overexpression. Western blots confirmed the overexpression of indicated proteins and VSV G proteins (MOI = 1.0, 12 h post-infection).

(B) Intracellular viral loads were significantly lower in DDX20-overexpressed cells than in the controls, as shown in fluorescence microscopy assay. Scale bars, 100 μ m.

(C) Viral RNA loads at 12 h post-infection were analyzed by measuring the virus G gene copy using qRT-PCR and normalized to the human beta-actin gene.

(D) Viral titration in Vero Cells was performed to identify the viral burdens of VSV in HEK293T cells.

(E) RNAi efficiency for DDX20 was shown in qRT-PCR analysis. Western blots demonstrated the interference efficiency of DDX20 and its effect on VSV-G protein expression.

(F) Intracellular viral loads of VSV-GFP in DDX20-silenced HEK293T cells were determined by monitoring GFP signal. Scale bars, 100 μ m.

(G) qRT-PCR analysis was used to identify the viral loads after silencing DDX20 in HEK293T cells.

(H) Viral titers in DDX20-silenced cells increased compared with that in control cells.

(I) DDX20 was overexpressed in HSV-1 infected cells.

(J) DDX20 overexpression inhibited the replication of the HSV-1 genome.

(K) DDX20 overexpression decreased the HSV-1 titer in the supernatant.

Results were expressed as the mean $\pm$ SEM. $**p < 0.01$, $***p < 0.001$, and $****p < 0.0001$ (t-test). Representative results from three independent experiments were shown.

and protein levels in HEK293T cells transfected with DDX20 plasmid, compared to the control group (Fig. 2A). Transient overexpression of DDX20 proteins led to a reduction in the expression of VSV G proteins (Fig. 2A) and viral-encoded GFP at 12 hpi (Fig. 2B), as well as a decrease in the levels of VSV RNA (Fig. 2C) and the release of infectious virus (Fig. 2D) in HEK293T cells.

To further validate the antiviral role of DDX20, we utilized two distinct shRNA plasmids to downregulate DDX20 expression in HEK293T cells. Both DDX20-specific shRNAs significantly decreased DDX20 mRNA and protein levels in transfected HEK293T cells, as evidenced by qPCR and Western blot analysis (Fig. 2E). Consistently, the knockdown of DDX20 led to a significant increase in the expression of VSV G proteins (Fig. 2E) and the viral-encoded GFP at 12 hpi (Fig. 2F), elevated levels of VSV RNA (Fig. 2G), and enhanced release of infectious virus (Fig. 2H) in HEK293T cells. These findings collectively support the conclusion that DDX20 acts to inhibit VSV replication.

To assess the impact of DDX20 on the replication of DNA viruses, HEK293T cells were infected with the DNA virus HSV-1. Following transfection of the DDX20-FLAG plasmid into HEK293T cells for 24 h (Fig. 2I), cells were subsequently infected with HSV-1 virus for 12 h. The RT-PCR assay and TCID₅₀ assay demonstrated that overexpression of DDX20 significantly inhibited the virus genome replication (Fig. 2J) and viral load (Fig. 2K). Collectively, these findings indicate that DDX20 exerts a substantial inhibitory effect on the replication of viruses such as VSV and HSV-1.

3.3. DDX20 regulates IFN- β production

Since type I interferons play pivotal roles in antiviral immune responses, we investigated whether DDX20 modulates the type I interferon pathway. To this end, we transfected HEK293T cells with DDX20-FLAG, followed by challenging the cells with VSV-GFP virus at a MOI of 1. The production of type I interferons, such as IFN- β , was assessed by qPCR. Remarkably, the overexpression of DDX20 significantly heightened the mRNA expression of IFN- β triggered by VSV (Fig. 3A), while knockdown of DDX20 with shRNAs resulted in a substantial reduction in the mRNA expression of IFN- β (Fig. 3B). These findings underscore the role of DDX20 in positively regulating the type I interferon pathway in response to viral challenges.

The RLR signaling pathway plays a pivotal role in the antiviral signaling cascade. RIG-I recognizes viral dsRNAs in the cytoplasm of virus-infected cells to induce downstream type I interferons response (Onomoto et al., 2021). To delineate the specific level at which DDX20 is involved in RIG-I signaling-mediated innate immune responses, we assessed the IFN- β promoter activity induced by the ectopic expression of key signaling molecules, including RIG-IN, MAVS, TBK1, and IRF3/5D. Notably, the overexpression of DDX20 significantly enhanced the IFN- β promoter activity triggered by RIG-IN, MAVS, TBK1, and IRF3/5D (Fig. 3C). Consistent with this, DDX20 overexpression markedly increased IFN mRNA expression induced by the ectopic expression of the N terminus of RIG-I (RIG-IN) (Fig. 3D). Conversely, DDX20 knockdown led to a significant suppression of the IFN- β promoter activity induced by RIG-IN, MAVS, TBK1, and IRF3/5D (Fig. 3E).

To further elucidate the impact of DDX20 on DNA viruses, we co-transfected cGAS and STING, the connector proteins of the DNA receptor signaling pathway, along with DDX20-FLAG to evaluate the activity of the IFN- β promoter. Our results revealed that DDX20 enhanced the activity of the IFN- β promoter triggered by cGAS/STING (Fig. 3F). In contrast, knockdown of DDX20 significantly attenuated the activity of the IFN- β promoter (Fig. 3G). These findings indicate that DDX20 positively modulates the promoter activity of IFN- β , thereby augmenting the expression of IFN- β in response to DNA viruses.

IFN- β exerts its antiviral effects by activating downstream ISRE elements through the JAK-STAT signaling pathway, thereby initiating the expression of interferon-stimulated genes (ISGs). To investigate the impact of DDX20 on ISRE promoter activity, we employed a luciferase

reporter approach. The results revealed that DDX20 knockdown inhibited the ISRE promoter activity activated by VSV (Fig. 3H). Moreover, DDX20 overexpression significantly augmented the activity of the ISRE luciferase reporter induced by RIG-IN, MAVS, TBK1, IKK ϵ , but not IRF3/5D (Fig. 3I). However, overexpression of DDX20 did not enhance ISRE promoter activity when stimulated with IFN- α and IFN- β , respectively (Fig. 3J). These findings suggest that DDX20 only contributes to the production of IFNs, thus indirectly promoting ISGs expression.

In addition, we observed that DDX20 altered the expression of SeV-induced interferon-stimulated genes (ISGs) such as IFIT1 and Cig5. The mRNA levels of these ISGs were significantly higher in DDX20-overexpressed cells but markedly lower in DDX20-silenced cells after SeV infection (Fig. 3K and L). These findings strongly suggest that DDX20 inhibits VSV replication by enhancing the production of type I interferons. Moreover, these results prompt further investigation into whether DDX20 functions by targeting IRF3, thereby serving as a positive regulator of RIG-I signaling.

3.4. DDX20 enhances IFN- β production by facilitating the interaction between IRF3 and TBK1

Given that induction of type I IFNs necessitates the activation of transcription factors such as IRF3 and NF- κ B, we sought to identify the transcription factors influenced by DDX20. To this end, we assessed IRF3-Luc and NF- κ B-Luc reporter activity using dual-luciferase reporter assay. We co-transfected the Renilla luciferase plasmid (pRL-TK) along with either the IRF3-Luciferase reporter plasmid or the NF- κ B-Luciferase reporter plasmid into DDX20 overexpressed cells, followed by stimulation with Sendai virus (SeV). Our result showed that the IRF3 promoter activity induced by SeV stimulation was significantly promoted by DDX20 overexpression but markedly inhibited upon DDX20 knockdown (Fig. 4A and B). However, DDX20 did not exert any discernible effect on the activity of the NF- κ B luciferase reporter triggered by SeV stimulation (Fig. 4C and D). These findings indicate that the augmentation of IFN- β promoter activity mediated by DDX20 is associated with the IRF3 signaling pathway rather than the NF- κ B signaling pathways.

IRF3 is the key transcription factor in the RIG-I pathway that promotes IFN- β expression. Upon RIG-I activation, activated TBK1 phosphorylates IRF3, leading to their homodimerization and translocation to the nucleus to bind to the IFN- β promoter (TNegishi et al., 2018). We next assessed the effect of DDX20 knockdown on the expression and phosphorylation of TBK1 and IRF3. Knockdown of DDX20 in SeV-stimulated cells resulted in a significant decrease in the phosphorylated IRF3 protein expression, while having no effect on the expression of IRF3, TBK1, or phosphorylated TBK1 protein (Fig. 4E).

IRF3 activation and subsequent IFN production require the assembly of a multi-protein complex involving TBK1 and IRF3. To investigate whether DDX20 interacts with IRF3 or TBK1, we performed reciprocal immunoprecipitation assays using HEK293T cells co-transfected with FLAG-tagged DDX20 and HA-tagged IRF3, or with HA-tagged DDX20 and FLAG-tagged TBK1. DDX20-FLAG or DDX20-HA were immunoprecipitated by anti-FLAG or anti-HA agarose, respectively. Both IRF3-HA and TBK1-FLAG were found in the DDX20 immunoprecipitants (Fig. 4F and G), suggesting that DDX20 can interact with TBK1 and IRF3. Furthermore, we examined whether DDX20 co-localizes with IRF3 or TBK1 in cells. Fluorescence microscopy revealed that DDX20 was localized to the nucleoplasm and cytosol. After co-overexpressing DDX20 with IRF3 or TBK1, colocalization analysis showed that DDX20 strongly colocalized with IRF3 in the cytoplasm of HeLa cells (Fig. 4H) and partially colocalized with TBK1 (Fig. 4I).

Having observed that the DDX20 protein can interact with TBK1 and IRF3, we investigated whether DDX20 plays a role in the formation of the TBK1-IRF3 complex, which is indispensable for IRF3 phosphorylation and IFN induction. To explore this, we co-transfected DDX20-shRNA and plasmids expressing TBK1-FLAG or IRF3-HA into HEK293T

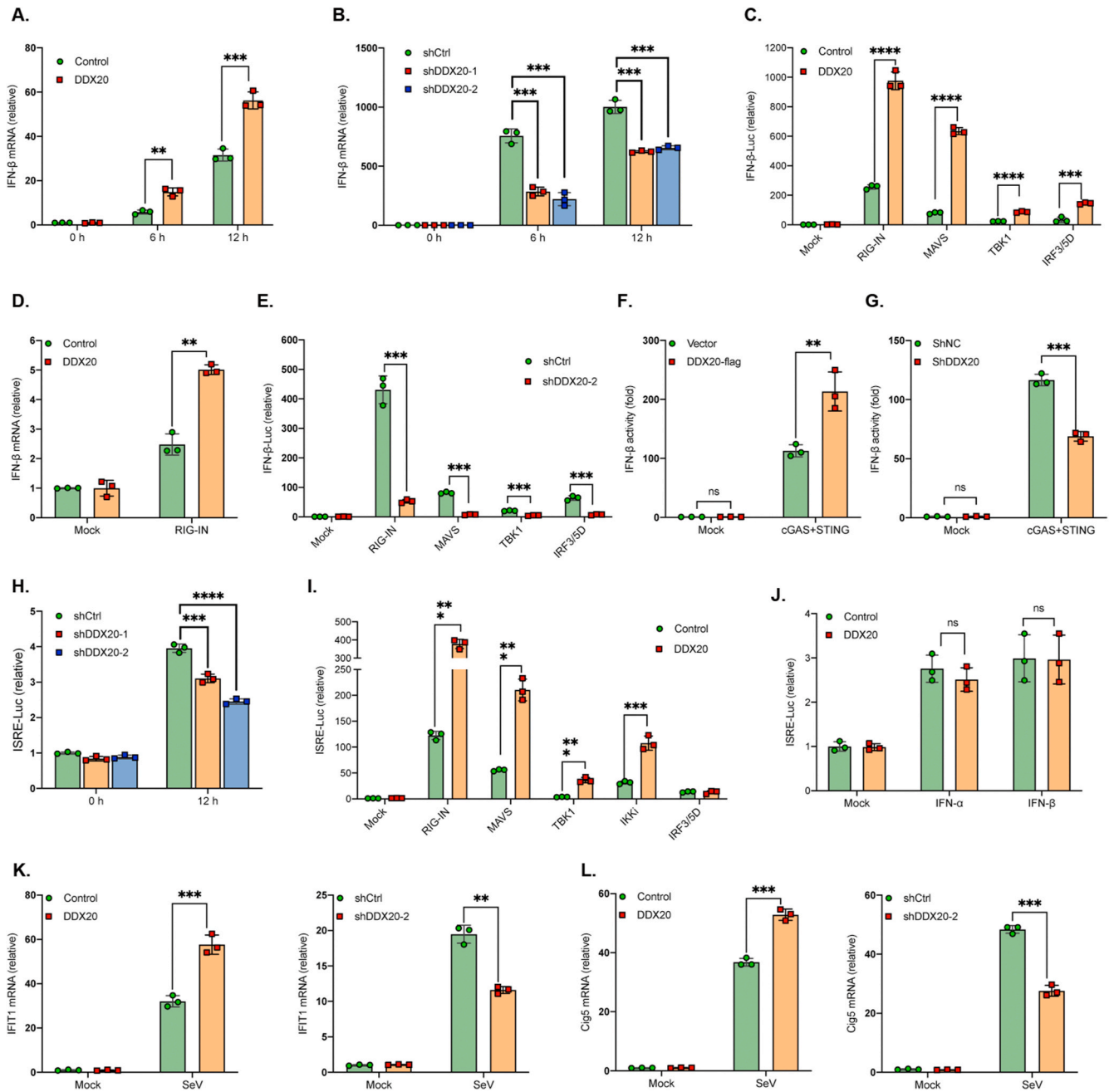


Fig. 3. DDX20 promotes IFN-β production in viral-infected HEK293T cells.

(A) Overexpressed DDX20 induced the expression of IFN-β during VSV infection.

(B) Knocking down DDX20 suppressed the expression of IFN-β during VSV infection.

(C) DLR assay showed that overexpression of DDX20 enhanced the IFN-β promoter activity induced by RIG-IN, MAVS, TBK1, IRF3/5D.

(D) qRT-PCR analysis showed that overexpression of DDX20 significantly increased IFN-β mRNA expression induced by ectopic expression of the N terminus of RIG-I (RIG-IN).

(E) DLR assay showed that knockdown of DDX20 significantly suppressed the IFN-β promoter activity induced by RIG-IN, MAVS, TBK1, and IRF3/5D.

(F) DLR assay showed that DDX20 overexpression enhanced the IFN-β promoter activity triggered by cGAS/STING.

(G) DLR assay showed that DDX20 knockdown reduced the IFN-β promoter activity triggered by cGAS/STING.

(H) 2fTGH cells with a luciferase reporter driven by the IFN-stimulated response element (ISRE) element (2fTGH-ISRE-luciferase cells) were used to measure the secretion of IFN proteins in the supernatants collected from VSV-infected cells after silencing DDX20.

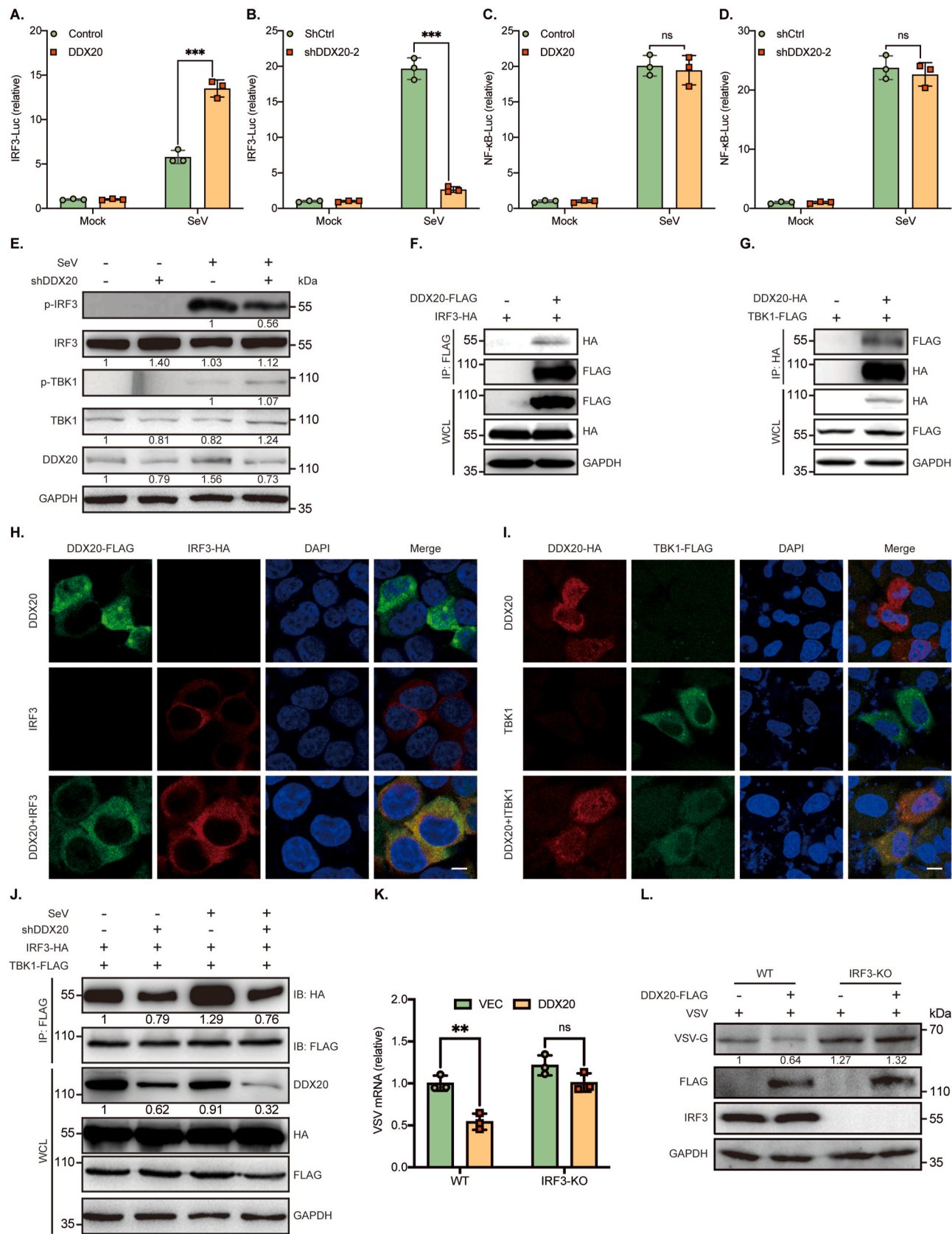
(I) Overexpression of DDX20 significantly augmented the activation of the ISRE luciferase reporter induced by RIG-IN, MAVS, TBK1, IKKε, but not IRF3/5D.

(J) DDX20 did not affect IFN-induced ISRE promoter activation.

(K) DDX20 overexpression increased the expression of SeV-induced IFIT1. Silencing DDX20 inhibited SeV-induced IFIT1 expression.

(L) DDX20 overexpression increased the expression of SeV-induced Cig5. Silencing DDX20 inhibited SeV-induced Cig5 expression.

** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$ (t -test). Representative results from three independent experiments are shown.



(caption on next page)

Fig. 4. DDX20 enhances IFN- β production by facilitating the interaction between IRF3 and TBK1

(A) DLR assay showed that overexpression of DDX20 promotes the IRF3 promoter activity induced by SeV stimulation.

(B) DLR assay showed that knockdown of DDX20 inhibits the IRF3 promoter activity induced by SeV stimulation.

(C–D) DDX20 did not affect the activity of the NF- κ B luciferase reporter induced by SeV stimulation.

(E) Knockdown of DDX20 significantly inhibited the phosphorylation level of IRF3 but had no significant effect on the phosphorylation level of TBK1.

(F) The exogenous interaction between DDX20 and IRF3 were detected by Co-IP assay.

(G) The exogenous interaction between DDX20 and TBK1 were detected by Co-IP assay.

(H) Confocal microscopy showed that DDX20 was co-localized with IRF3 in HeLa cells. Scale bars, 10 μ m.(I) Confocal microscopy showed that DDX20 was partially colocalized with TBK1 in HeLa cells. Scale bars, 10 μ m.

(J) Knockdown of DDX20 reduced the interaction between TBK1 and IRF3.

(K–L) DDX20 does not affect the expression of viral mRNA (K) or VSV-G protein (L) in IRF3-KO cell lines at 12 h post-infection.

* $p < 0.05$, and *** $p < 0.001$ (t-test). Representative results from three independent experiments are shown.

cells, followed by SeV stimulation. Subsequently, Co-IP using anti-FLAG antibodies demonstrated that knockdown of DDX20 reduced the interaction between TBK1-FLAG and IRF3-HA (Fig. 4J), suggesting that the DDX20 is involved in the formation of the TBK1-IRF3 complex.

Finally, we investigated whether DDX20 retained its ability to inhibit viral growth in IRF3 knockout (KO) cells. Our previous results showed that DDX20 overexpression reduced both the levels of VSV RNA and VSV-G protein; however, such an inhibitory effect was absent in IRF3-deficient HEK293T cells (Fig. 4K and L), indicating that the antiviral effect of DDX20 is IRF3-dependent. Thus, we concluded that DDX20 impedes VSV propagation by interacting with IRF3 to activate type I IFN production in HEK293T cells.

3.5. DDX20 gene-deficient mice exhibited increased susceptibility to viral infection

Because *Ddx20* is essential for early embryonic development and its complete loss is fatal in mice (Mouillet et al., 2008), we generated *Ddx20* heterozygous mice to study the biological function of DDX20 *in vivo*. Heterozygous mice (*Ddx20*^{+/-}) were bred to generate *Ddx20*^{+/-} mice and *Ddx20*-sufficient control littermates (*Ddx20*^{+/+}). Subsequently, eight-week-old *Ddx20*^{+/+} and *Ddx20*^{+/-} mice were intranasally infected with VSV, and their body weights and survival curve were monitored. *Ddx20*^{+/-} mice exhibited a significant weight loss post infection compared to *Ddx20*^{+/+} mice (Fig. 5A). Survival analysis showed that >30% (7 of 21) of *Ddx20*^{+/-} mice succumbed to intranasal VSV infection at a dose of 5×10^5 PFU, whereas only 1 out of 21 *Ddx20*^{+/+} mice died (Fig. 5A), indicating that *Ddx20*^{+/-} mice were more susceptible to intranasal VSV infection. Consistently, viral burden in the serum of *Ddx20*^{+/-} mice was significantly higher than that in *Ddx20*^{+/+} mice on day 1 post-infection (Fig. 5B). We observed that DDX20 expression in the brain, kidney, liver, lung, and spleen from uninfected *Ddx20*^{+/-} mice was significantly lower than in *Ddx20*^{+/+} mice (Fig. 5C). Moreover, the viral load in multiple tissues (brain, kidney, liver, lung, and spleen) from *Ddx20*^{+/-} mice on the third day post-infection was also significantly higher than that of control mice (Fig. 5D–F). Correspondingly, histological examination revealed much more severe VSV-infection-induced structural damage and increased inflammatory cell infiltration in the lung and brain of *Ddx20*^{+/-} mice compared to wild-type littermates (Fig. 5G).

Type I interferon plays a pivotal role in protecting mice from VSV infection (Muller et al., 1994). To assess the systemic role of *Ddx20* in type I IFN production, we measured *Ifn β* mRNA levels in VSV-infected *Ddx20*^{+/-} mice. As expected, *Ifn β* mRNA levels were suppressed in the tissues of *Ddx20*^{+/-} mice on the third day post-infection (Fig. 6A–F). Taken together, these findings support the notion that downregulation of DDX20 inhibits type I interferon production, thereby facilitating VSV replication *in vivo*.

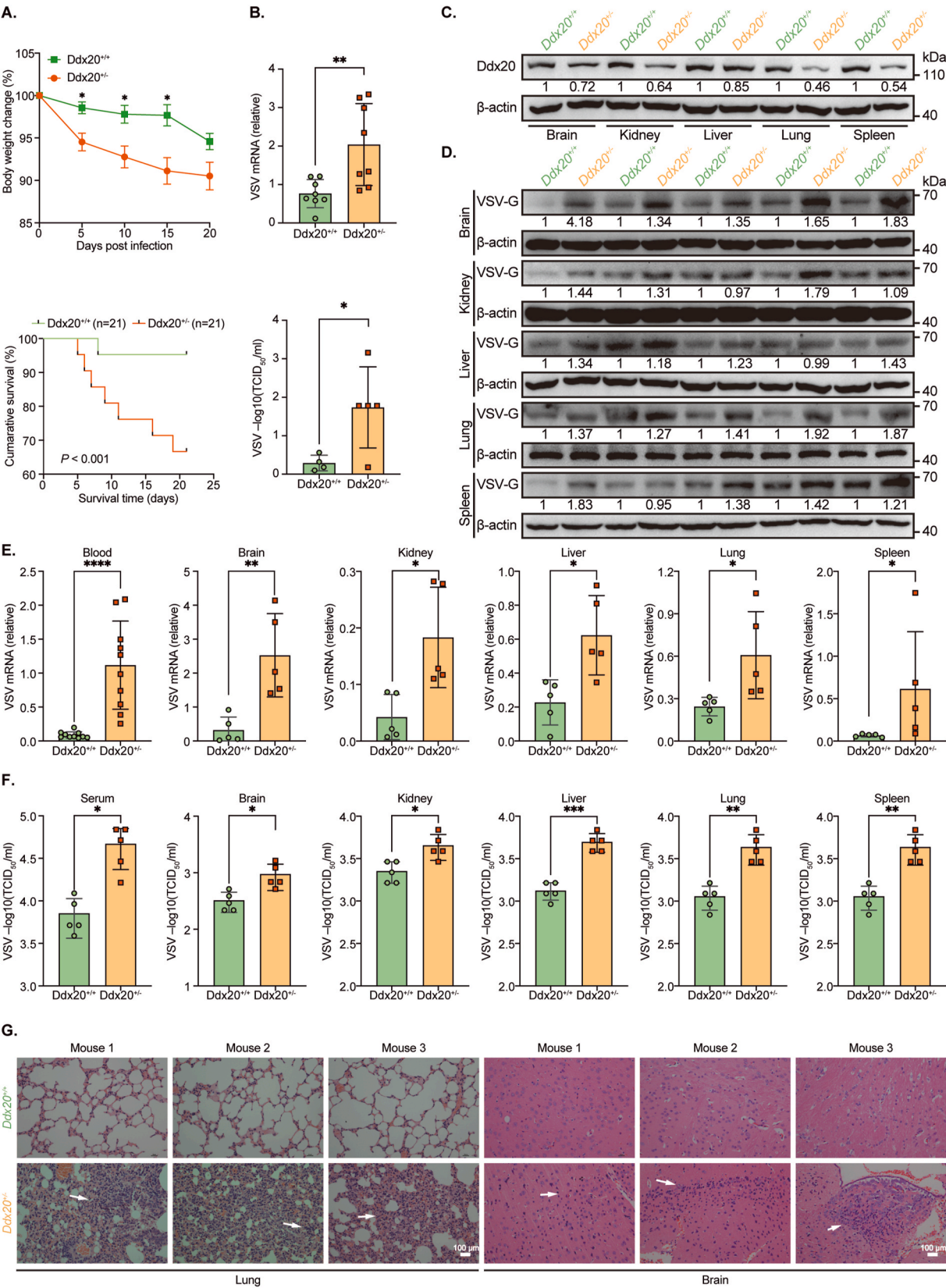
4. Discussion

DEAD-box RNA helicases constitute a major subfamily of RNA helicases, playing crucial roles in various RNA metabolic processes,

including RNA unwinding, RNA processing, and RNPase activities. While certain DDXs, such as DDX2A1 (also known as eukaryotic translation initiation factor 4A1, eIF4A1), DDX2B (eIF4A2), DDX48 (IF4AIII), DDX3 (the yeast ortholog of Ded1), and DDX19 (the yeast homolog of Dbp5), have well-defined physiological functions, the roles of most DDXs remain poorly understood (Andrisani et al., 2022; Linder and Jankowsky, 2011). Some DDXs have been implicated in human diseases, particularly viral infections (Ali, 2021b; Baldaccini and Pfeffer, 2021) and cancer (Robert and Pelletier, 2013; van Voss et al., 2017). Our study focuses on DDX20, a prominent member of the DEAD-box RNA helicase family, to unravel its diverse roles in suppressing viral infections, promoting immune responses, and potentially influencing interferon pathway. These findings collectively underscore the significance of DDX20 as a pivotal antiviral regulator and a potential target for the development of antiviral drugs.

Research have demonstrated that the cytoplasmic protein RLRs (RIG-I, MDA5, and LGP2) primarily recognize viral nucleic acid components within the cell. These receptors discern various RNA structures, thereby facilitating the detection of diverse viruses. Recent investigations have increasingly elucidated the involvement of DEAD-box helicase family members in the host's innate immune response to viral infections. For instance, RIG-I/DDX58 operates as a recognition receptor within the RLR signaling pathway, triggering the synthesis of type I interferons and cytokines (Kato et al., 2006). Another member, DDX41 serves as a DNA sensor that utilizes STING to detect pathogenic DNA. It modulates dendritic cell responses to DNA viruses and induces the production of type I interferons and antiviral cytokines (Zhang et al., 2011). These insights underscore the diverse roles of DEAD-box helicases in safeguarding the host against viral challenges. Both DDX3 and DDX3X can promote the activation of IFN- β activated by TBK1/IKK ϵ (Soulat et al., 2008). DDX3X, as a substrate of TBK1 kinase, is phosphorylated upon stimulation, then binds to the IFN- β promoter to augment its expression (Li et al., 2015; Soulat et al., 2008). Virus infection results in the association of DDX3 with IKK ϵ , thereby enhancing the promoter activity of IFN- β (Soulat et al., 2008). Besides, DDX3 can regulate the production of interferon IFN- β by interacting with TBK1, enhancing the expression of NF- κ B (Cartwright et al., 2018). Our previous work has indicated that DDX25 inhibits the production of IFN- β through both the IRF3 and NF- κ B pathways, promoting the replication of dengue virus. Specifically, DDX25 inhibits the transcriptional activity of the IFN- β promoter induced by overexpression of RIG-I, MDA5, MAVS, TBK1, and IRF3. At the same time, DDX25 also inhibits the transcriptional activity of NF- κ B induced by overexpression of RIG-I, IKK α , and p65 subunits (Feng et al., 2017).

DDX20, despite being a member of the DEAD-box helicase family, exhibits multifunctionality beyond its canonical role as a helicase during transcription. Instead, DDX20 functions as an auxiliary factor in both transcriptional activation and inhibition (Mouillet et al., 2008; Shpargel et al., 2009). Notably, DDX20 has been identified as an inhibitor of Steroidogenic Factor 1 Nuclear Receptor (SF1) transcription. In this regulatory process, a non-conservative structure located at the C-terminus of DDX20 assumes a pivotal role. This structural feature enables DDX20 to interact with SF1, thereby exerting its inhibitory influence on



(caption on next page)

Fig. 5. DDX20 gene-deficient mice are more susceptible to viral infection

(A) Changes in body weight and survival rate post-VSV infection. Data are expressed as mean $\pm$ SEM.

(B) Viral burdens in mouse blood on day 1 post-VSV infection. The VSV-G gene copy number was measured by qRT-PCR. Viral titer in the serum of mice was detected with TCID₅₀ assay using Vero Cells.

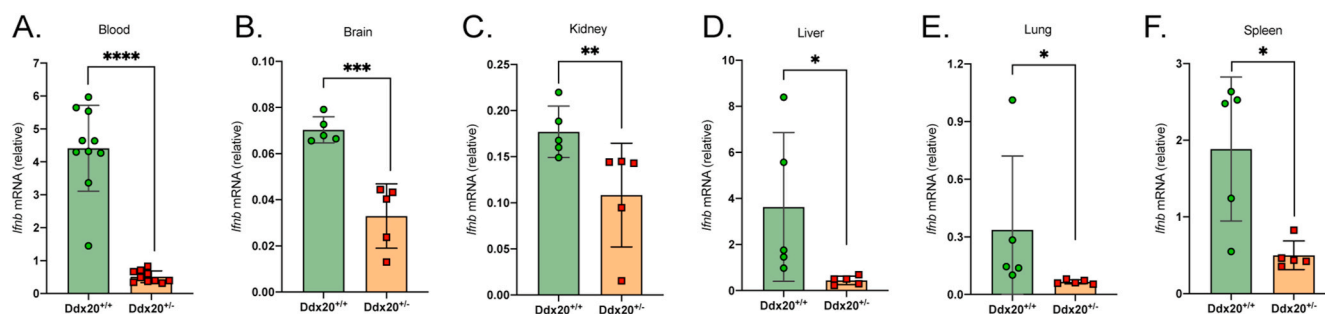
(C) Depletion of *Ddx20* in uninfected C57BL/6J *Ddx20* heterozygous mice was confirmed by Western blot analysis of *Ddx20* expression in mouse brain, kidney, liver, lung, and spleen.

(D) Western blot analysis of the VSV-G protein in mouse tissues (brain, kidney, liver, lung, and spleen) on day 3 post-infection.

(E–F) The VSV-G gene copy number (E) and viral titer (F) in mouse blood and tissues (brain, kidney, liver, lung, and spleen) on day 3 post-VSV infection were detected by qRT-PCR and TCID₅₀ assay, respectively. Data in (E) ‘blood’ were pooled from two independent experiments.

(G) H&E staining of lung and brain tissues of *Ddx20*^{+/-} mice at day 5 post-VSV infection. VSV-infection-induced structural damage and inflammatory cell infiltration were indicated by white arrows. Scale bars, 100 μ m

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$, (t-test). Representative results from three independent experiments are shown.

**Fig. 6.** *Ddx20*^{+/-} mice had impaired IFN- β during viral infection.

(A–F) qRT-PCR analysis of the *Ifnb* mRNA levels from tissue samples (blood, brain, kidney, liver, lung, and spleen) from VSV-infected *Ddx20*^{+/-} mice. Data from two pooled independent experiments (A) or one representative of three independent experiments (B–F) are shown (n = 5 mice per group).

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$, (t-test).

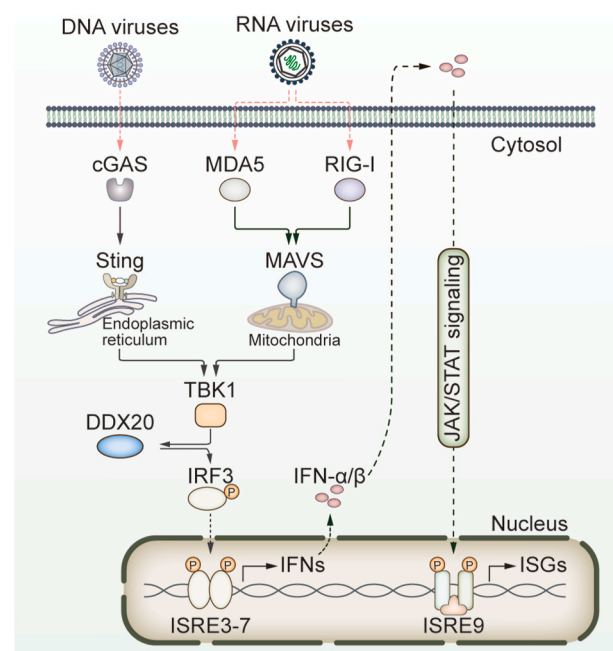
SF1 transcription.

Moreover, research findings indicate that DDX20 is indispensable for embryonic development and ovarian structure in mice. Its impact extends to the regulation of sex hormone secretion (Mouillet et al., 2008). In the context of movement-related functions, DDX20 plays a significant role in the motor function and pupation process of *Drosophila* larvae (Shpargel et al., 2009). This multifaceted involvement of DDX20 underscores its diverse roles in cellular processes, extending beyond its traditional classification as a helicase.

Viruses have evolved various mechanisms to antagonize host immune responses, enabling them to evade detection and clearance by the host's immune system. The analysis of microarray-based gene expression data from the GEO database (GSE20948, GSE3397, and GSE67198) indicated a consistent pattern of DDX20 mRNA downregulation in cells infected with various viruses compared to mock-treated cells at 24 hpi or 48 hpi (Fig.S1 A–C). This observation was further supported by real-time quantitative PCR (qPCR) and Western blot analysis conducted on HeLa and HEK293T cells infected with VSV or SeV (Fig.S1 D–G). These findings suggest that DDX20 expression is transcriptionally suppressed in response to viral infection, implying a potential role for DDX20 as a host defense factor but is downregulated by viruses.

5. Conclusion

This study offers valuable insights into the antiviral function of DDX20, emphasizing its pivotal role in orchestrating the innate immune response. DDX20 accomplishes this by enhancing interferon expression and fine-tuning the interaction between TBK1 and IRF3, thereby impeding viral replication. The findings of this research significantly contribute to a more profound comprehension of immune responses and pave the way for exploring novel avenues in antiviral therapeutics. Based on our discoveries, we propose the following working model (Fig. 7) to elucidate how DDX20 positively regulates the production of type I interferons in the antiviral innate response.

**Fig. 7.** Working model of how DDX20 positively regulates the production of type I interferons in the antiviral innate response. During virus infection, DDX20 can interact with TBK1 and IRF3 and promotes IRF3 activation, enhancing the production of IFNs, which in turn facilitate transcriptional activation of ISGs.

CRediT authorship contribution statement

Zhiqiang Chen: Writing – original draft, Methodology, Data curation, Conceptualization. **Jinyu Zhang:** Methodology, Data curation. **Tingting Feng:** Writing – review & editing, Writing – original draft,

Project administration, Data curation, Conceptualization. **Xiujuan Wang:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shimeng Zhou:** Methodology, Data curation, Conceptualization. **Wen Pan:** Resources, Project administration, Methodology. **Zhengrong Chen:** Supervision, Resources. **Yongdong Yan:** Supervision, Resources. **Jianfeng Dai:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The manuscript includes all datasets generated or analyzed during this study.

Acknowledgements

We would like to thank Dr. Bo Hu for technical assistance. This research was supported by grants from the Priority Academic Program Development of Jiangsu Higher Education Institutions, National Natural Science Foundation of China (32170142 and 81971917), Jiangsu Natural Science Foundation (BK20211310), Basic Frontier Innovation Cross Program of Suzhou Medical College (MP24906523) and fundings from Suzhou International Joint Laboratory for Diagnosis and Treatment of Brain Diseases.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.antiviral.2024.105875>.

References

- Alexopoulou, L., Holt, A.C., Medzhitov, R., Flavell, R.A., 2001. Recognition of double-stranded RNA and activation of NF- κ B by Toll-like receptor 3. *Nature* 413, 732–738. <https://doi.org/10.1038/35099560>.
- Ali, M.A.M., 2021a. The DEAD-box protein family of RNA helicases: sentinels for a myriad of cellular functions with emerging roles in tumorigenesis. *Int. J. Clin. Oncol.* 26, 795–825. <https://doi.org/10.1007/s10147-021-01892-1>.
- Ali, M.A.M., 2021b. DEAD-box RNA helicases: the driving forces behind RNA metabolism at the crossroad of viral replication and antiviral innate immunity. *Virus Res.* 296, 198352. <https://doi.org/10.1016/j.virusres.2021.198352>.
- Andrisani, O., Liu, Q., Kehn, P., Leitner, W.W., Moon, K., Vazquez-Maldonado, N., Fingerman, I., Gale, M., 2022. Biological functions of DEAD/DEAH-box RNA helicases in health and disease. *Nat. Immunol.* 23, 354–357. <https://doi.org/10.1038/s41590-022-01149-7>.
- Anwar, S., Ul Islam, K., Azmi, M.I., Iqbal, J., 2021. cGAS-STING-mediated sensing pathways in DNA and RNA virus infections: crosstalk with other sensing pathways. *Arch. Virol.* 166, 3255–3268. <https://doi.org/10.1007/s00705-021-05211-x>.
- Baldacchini, M., Pfeffer, S., 2021. Untangling the roles of RNA helicases in antiviral innate immunity. *PLoS Pathog.* 17, e1010072. <https://doi.org/10.1371/journal.ppat.1010072>.
- Cai, Q., Guo, Y., Xiao, B., Banerjee, S., Saha, A., Lu, J., Glisovic, T., Robertson, E.S., 2011. Epstein-Barr virus nuclear antigen 3C stabilizes Gemin3 to block p53-mediated apoptosis. *PLoS Pathog.* 7, e1002418. <https://doi.org/10.1371/journal.ppat.1002418>.
- Cartwright, T.N., Worrell, J.C., Marchetti, L., Dowling, C.M., Knox, A., Kiely, P., Mann, J., Mann, D.A., Wilson, C.L., 2018. HDAC1 interacts with the p50 NF- κ B subunit via its nuclear localization sequence to constrain inflammatory gene expression. *Biochim Biophys Acta Gene Regul Mech* 1861, 962–970. <https://doi.org/10.1016/j.bbargm.2018.09.001>.
- Diebold, S.S., Kaisho, T., Hemmi, H., Akira, S., Reis e Sousa, C., 2004. Innate antiviral responses by means of TLR7-mediated recognition of single-stranded RNA. *Science* 303, 1529–1531. <https://doi.org/10.1126/science.1093616>.
- Feng, T., Sun, T., Li, G., Pan, W., Wang, K., Dai, J., 2017. DEAD-box helicase DDX25 is a negative regulator of type I interferon pathway and facilitates RNA virus infection. *Front. Cell. Infect. Microbiol.* 7, 356. <https://doi.org/10.3389/fcimb.2017.00356>.
- Fuller-Pace, F.V., 2006. DEXD/H box RNA helicases: multifunctional proteins with important roles in transcriptional regulation. *Nucleic Acids Res.* 34, 4206–4215. <https://doi.org/10.1093/nar/gkl460>.
- Grundhoff, A.T., Kremmer, E., Tureci, O., Glieden, A., Gindorf, C., Atz, J., Mueller-Lantzsch, N., Schubach, W.H., Grässer, F.A., 1999. Characterization of DP103, a novel DEAD box protein that binds to the Epstein-Barr virus nuclear proteins EBNA2 and EBNA3C. *J. Bio. Chem.* 274 (27), 19136–19144. <https://doi.org/10.1074/jbc.274.27.19136>.
- Han, P., Ye, W., Lv, X., Ma, H., Weng, D., Dong, Y., Cheng, L., Chen, H., Zhang, L., Xu, Z., Lei, Y., Zhang, F., 2017. DDX50 inhibits the replication of dengue virus 2 by upregulating IFN- β production. *Arch. Virol.* 162, 1487–1494. <https://doi.org/10.1007/s00705-017-3250-3>.
- Heil, F., Hemmi, H., Hochrein, H., Ampenberger, F., Kirschning, C., Akira, S., Lipford, G., Wagner, H., Bauer, S., 2004. Species-specific recognition of single-stranded RNA via toll-like receptor 7 and 8. *Science* 303, 1526–1529. <https://doi.org/10.1126/science.1093620>.
- Kato, H., Takeuchi, O., Sato, S., Yoneyama, M., Yamamoto, M., Matsui, K., Uematsu, S., Jung, A., Kawai, T., Ishii, K.J., Yamaguchi, O., Otsu, K., Tsujimura, T., Koh, C.S., Reis e Sousa, C., Matsuura, Y., Fujita, T., Akira, S., 2006. Differential roles of MDA5 and RIG-I helicases in the recognition of RNA viruses. *Nature* 441, 101–105. <https://doi.org/10.1038/nature04734>.
- Li, G.H., Feng, T.T., Pan, W., Shi, X.H., Dai, J.F., 2015. DEAD-box RNA helicase DDX3X inhibits DENV replication via regulating type one interferon pathway. *Biochem Biophys Res Commun* 456, 327–332. <https://doi.org/10.1016/j.bbrc.2014.11.080>.
- Linder, P., Jankowsky, E., 2011. From unwinding to clamping - the DEAD box RNA helicase family. *Nat. Rev. Mol. Cell Biol.* 12, 505–516. <https://doi.org/10.1038/nrm3154>.
- Loo, Y.M., Gale Jr., M., 2011. Immune signaling by RIG-I-like receptors. *Immunity* 34, 680–692. <https://doi.org/10.1016/j.immuni.2011.05.003>.
- Miao, Y., Zhang, T., Guan, M., Zhao, Q., Zhang, R., Liu, X., Ma, T., Ren, T., Zheng, Z., He, W., Tian, W., Cui, Q., Zhai, X., Zuo, Y., Zhu, H., Zheng, H., Yuan, Y., 2024. DDX4 enhances antiviral activity of type I interferon by disrupting interaction of USP7/SOCS1 and promoting degradation of SOCS1. *mBio* 15, e0321323. <https://doi.org/10.1128/mbio.03213-23>.
- Miyashita, M., Oshiumi, H., Matsumoto, M., Seya, T., 2011. DDX60, a DEXD/H box helicase, is a novel antiviral factor promoting RIG-I-like receptor-mediated signaling. *Mol. Cell Biol.* 31, 3802–3819. <https://doi.org/10.1128/MCB.01368-10>.
- Mouillet, J.F., Yan, X., Ou, Q., Jin, L., Muglia, L.J., Crawford, P.A., Sadovsky, Y., 2008. DEAD-box protein-103 (DP103, Ddx20) is essential for early embryonic development and modulates ovarian morphology and function. *Endocrinology* 149, 2168–2175. <https://doi.org/10.1210/en.2007-1237>.
- Muller, U., Steinhoff, U., Reis, L.F., Hemmi, S., Pavlovic, J., Zinkernagel, R.M., Aguet, M., 1994. Functional role of type I and type II interferons in antiviral defense. *Science* 264, 1918–1921. <https://doi.org/10.1126/science.8009221>.
- Onomoto, K., Onoguchi, K., Yoneyama, M., 2021. Regulation of RIG-I-like receptor-mediated signaling: interaction between host and viral factors. *Cell. Mol. Immunol.* <https://doi.org/10.1038/s41423-020-00602-7>.
- Oshiumi, H., Sakai, K., Matsumoto, M., Seya, T., 2010. DEAD/H BOX 3 (DDX3) helicase binds the RIG-I adaptor IPS-1 to up-regulate IFN- β -inducing potential. *Eur. J. Immunol.* 40, 940–948. <https://doi.org/10.1002/eji.200940203>.
- Robert, F., Pelletier, J., 2013. Perturbations of RNA helicases in cancer. *Wires Rna* 4, 333–349. <https://doi.org/10.1002/wrna.1163>.
- Schroder, M., Baran, M., Bowie, A.G., 2008. Viral targeting of DEAD box protein 3 reveals its role in TBK1/IKK ϵ -mediated IRF activation. *EMBO J.* 27, 2147–2157. <https://doi.org/10.1038/emboj.2008.143>.
- Shpargel, K.B., Praveen, K., Rajendra, T.K., Matera, A.G., 2009. Is an essential gene required for larval motor function and pupation in. *Mol. Biol. Cell* 20, 90–101. <https://doi.org/10.1091/mbc.E08-01-0024>.
- Soulad, D., Bürcckstümmer, T., Westermayer, S., Gonçalves, A., Bauch, A., Stefanovic, A., Hantschel, O., Bennett, K.L., Decker, T., Superti-Furga, G., 2008. The DEAD-box helicase DDX3X is a critical component of the TANK-binding kinase 1-dependent innate immune response. *EMBO J.* 27, 2135–2146. <https://doi.org/10.1038/emboj.2008.126>.
- TNegishi, H., Taniguchi, T., Yanai, H., 2018. The interferon (IFN) class of cytokines and the IFN regulatory factor (IRF) transcription factor family. *Cold Spring Harbor Perspect. Biol.* 10 (11), a028423. <https://doi.org/10.1101/cshperspect.a028423>.
- van Voss, M.R.H., van Diest, P.J., Raman, V., 2017. Targeting RNA helicases in cancer: the translation trap. *Bba-Rev Cancer* 1868, 510–520. <https://doi.org/10.1016/j.bbcan.2017.09.006>.
- Zhang, K., Zhang, Y., Xue, J., Meng, Q., Liu, H., Bi, C., Li, C., Hu, L., Yu, H., Xiong, T., Yang, Y., Cui, S., Bu, Z., He, X., Li, J., Huang, L., Weng, C., 2019. DDX19 inhibits type I interferon production by disrupting TBK1-IRK ϵ -IRF3 interactions and promoting TBK1 and IKK ϵ degradation. *Cell Rep.* 26, 1258–1272. <https://doi.org/10.1016/j.celrep.2019.01.029>.
- Zhang, Y., Cen, J., Yuan, G., Jia, Z., Chen, K., Gao, W., Chen, J., Adamek, M., Jia, Z., Zou, J., 2023. DDX5 inhibits type I IFN production by promoting degradation of TBK1 and disrupting formation of TBK1 - TRAF3 complex. *Cell. Mol. Life Sci.* 80, 212. <https://doi.org/10.1007/s00018-023-04860-2>.
- Zhang, Z.Q., Yuan, B., Bao, M.S., Lu, N., Kim, T., Liu, Y.J., 2011. The helicase DDX41 senses intracellular DNA mediated by the adaptor STING in dendritic cells. *Nat. Immunol.* 12, 959–962. <https://doi.org/10.1038/ni.2091>.

- Zhao, T.J., Yang, L., Sun, Q., Arguello, M., Ballard, D.W., Hiscott, J., Lin, R.T., 2007. The NEMO adaptor bridges the nuclear factor-kappa B and interferon regulatory factor signaling pathways. *Nat. Immunol.* 8, 592–600. <https://doi.org/10.1038/ni1465>.
- Zhong, B., Yang, Y., Li, S., Wang, Y.Y., Li, Y., Diao, F., Lei, C., He, X., Zhang, L., Tien, P., Shu, H.B., 2008. The adaptor protein MITA links virus-sensing receptors to IRF3 transcription factor activation. *Immunity* 29, 538–550. <https://doi.org/10.1016/j.immuni.2008.09.003>.
- Zhou, Y., Wu, W., Xie, L., Wang, D., Ke, Q., Hou, Z., Wu, X., Fang, Y., Chen, H., Xiao, S., Fang, L., 2017. Cellular RNA helicase DDX1 is involved in transmissible gastroenteritis virus nsp14-induced interferon-beta production. *Front. Immunol.* 8, 940. <https://doi.org/10.3389/fimmu.2017.00940>.